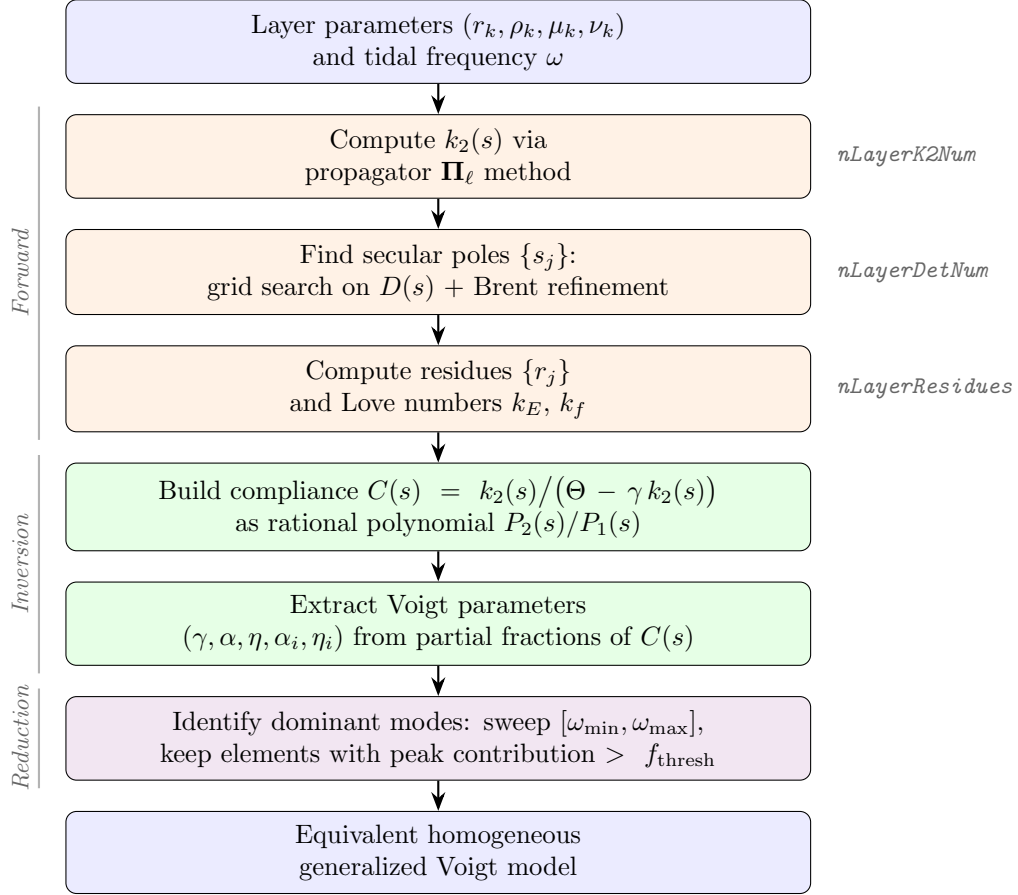


Algorithm: Multilayer to Homogeneous Voigt Rheology



References. *Gevorgyan, Matsuyama & Ragazzo (2023), MNRAS 523, 1822.*

Sabadini, Vermeersen & Cambiotti (2016), Global Dynamics of the Earth, 2nd ed., Springer.

Full algorithm

Notation

Layered-body quantities follow Sabadini, Vermeersen & Cambiotti (2016): μ_k shear modulus, ν_k viscosity, $\mathbf{Y}_\ell$ fundamental matrix, $\mathbf{\Pi}_\ell$ propagator, $\mathbf{I}_C$ core boundary matrix, $D(s)$ secular determinant. Homogeneous-body quantities follow Gevorgyan, Matsuyama & Ragazzo (2023, hereafter GMR 2023): $C(s)$ compliance, $P_1(s)/P_2(s)$ polynomials, γ , α , η , α_i , η_i Voigt parameters.

Input

Layer parameters $\{r_k, \rho_k, \mu_k, \nu_k\}_{k=1}^N$ (center to surface), tidal frequency ω , working precision p , pole-search grid bounds $[a_{\min}, a_{\max}]$ with G points, frequency interest range $[\omega_{\min}, \omega_{\max}]$, dominant threshold f_{thresh} .

Stage 1: Forward problem

1. Pre-process the input: any run of consecutive liquid layers ($\mu_k < 1$ Pa) is collapsed into a single equivalent liquid layer with the outer radius of the outermost member, the inner radius of the underlying solid, and the volume-weighted mean density

$$\rho_{\text{eff}} = \frac{\sum_{k=a}^b \rho_k (r_k^3 - r_{k-1}^3)}{r_b^3 - r_{a-1}^3},$$

which preserves the total mass of the merged region exactly. This step is a no-op for inputs with at most one liquid run; it removes spurious internal interfaces when a stratified liquid core or mantle is supplied as several sub-layers.

2. For each remaining layer k , form the Maxwell complex shear modulus via the Correspondence Principle: $\hat{\mu}_k(s) = \mu_k \nu_k s / (\mu_k + \nu_k s)$. For liquid layers ($\mu_k < 1$ Pa), regularize $\hat{\mu}_k \approx 10^{-20}$ Pa.
3. Build the 6×6 fundamental matrix $\mathbf{Y}_\ell(r)$ for each layer (degree $\ell = 2$; Sabadini et al. 2016, Eq. 2.42), where each column is an independent solution of the spheroidal system within that layer.
4. Form the propagator $\mathbf{\Pi}_\ell(r, r') = \mathbf{Y}_\ell(r) \mathbf{Y}_\ell^{-1}(r')$ (Eq. 2.9) and chain through all N layers from the core-mantle boundary matrix $\mathbf{I}_C$ (Eq. 2.6) to the surface.
5. Compute the secular determinant $D(s) = \det(\mathbf{P}_1 \mathbf{\Pi}_\ell(R, r_C) \mathbf{I}_C) \big|_{\mu=\hat{\mu}(s)}$ (Eq. 1.199) and extract $k_2(s)$ from the surface boundary conditions.

Stage 2: Modal decomposition

6. Evaluate $k_E = \text{Re}[k_2(10^{10})]$ (elastic limit) and $k_2(i\omega)$ at the tidal frequency.
7. Build a logarithmic grid $s_n = -10^{a_n}$ with G points in $[a_{\min}, a_{\max}]$. Evaluate $D(s_n)$ at each grid point.
8. Detect sign changes in $D(s)$ between consecutive grid points. Refine each bracket with Brent's method to obtain all M secular poles $\{s_j\}$, $s_j < 0$ (Sabadini et al. 2016, Eq. 1.200).
9. For each pole, compute the residue via the limit formula: $r_j = \delta \cdot k_2(s_j + \delta)$ with $\delta = 10^{-6}|s_j|$.
10. Compute the fluid Love number $k_f = k_E + \sum_{j=1}^M (-r_j/s_j)$. The partial-fraction form is (GMR 2023, Eq. 26): $k_2(s) = k_E + \sum_{j=1}^M r_j/(s - s_j)$.

Stage 3: Rheological inversion

11. Compute mapping constants (GMR 2023, Eqs. 2, 4): $\Theta = 3I_o G/R^5$, $\gamma = \Theta/k_f$, $\alpha = \Theta/k_E - \gamma$.
12. Build the compliance as a rational polynomial (GMR 2023, Eq. 6): $C(s) = P_2(s)/P_1(s) = k_2(s)/(\Theta - \gamma k_2(s))$, using the partial-fraction form of k_2 .
13. Factor the $s = 0$ root from $P_1(s)$ to extract the isolated dashpot parameter η .
14. Solve $P_2(s) = 0$ for the $N_V = M - 1$ Voigt poles $\{q_i\}$, $q_i < 0$ (GMR 2023, Eq. 20).
15. For each Voigt pole, extract the spring and dashpot compliance parameters α_i, η_i from the partial-fraction coefficients of $C_v(s)$ (Eq. 23); characteristic time $\omega_i^{-1} = \eta_i/\alpha_i$ (Eq. 18).
16. Convert from compliance space to physical units: multiply $\gamma, \alpha, \eta, \alpha_i, \eta_i$ by $F = 5\bar{\rho}R^2/38$.

Stage 4: Dominant mode selection

17. Sweep 200 log-spaced frequencies in $[\omega_{\min}, \omega_{\max}]$.
18. At each frequency, compute each Voigt element's fractional contribution to $-\text{Im}[C(i\omega)]$.
19. Record the maximum fractional contribution of each element across the sweep.
20. Retain element i if $\max_{\omega}(-\text{Im}[C_i]/-\text{Im}[C]) > f_{\text{thresh}}$.
21. Build the reduced model from the spring $1/\alpha$, the dashpot $1/(\eta s)$, and the retained Voigt elements.

Output

- Love numbers: $k_E, k_f, k_2(i\omega)$.
- Full Voigt model: $\gamma, \alpha, \eta, \{(\alpha_i, \eta_i, \omega_i^{-1})\}_{i=1}^{N_V}$.
- Secular poles and residues: $\{(s_j, r_j)\}_{j=1}^M$.
- Reduced Voigt model: dominant elements only, with k_2 comparison and relative errors.
- Comparison plots: $\text{Re}[k_2]$ and $-\text{Im}[k_2]$ vs ω for full and reduced models.